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Palaeomagnetism of a Pan-African diorite: a Late Precambrian pole for western Africa

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Summary. A study of a Late Pan-African diorite (Adma diorite; 18.3°N, 1.2° E) gives a mean direction of magnetization $D = 317.3°$, $I = 78.6°$ (13 sites, 62 samples, $\alpha_{95} = 9.0°$) based on alternating field and thermal cleaning. The corresponding palaeopole is at 32.5° N, 344.7° E ($A_{95} = 15.9°$). The diorite is dated by U/Pb method at 616 ± 11 Myr. The age of this magnetization is believed to be somewhat younger, between 590 and 616 Myr. Thermal treatment shows that above 675°C (Curie point of synthetic hematite and the highest value usually observed in rocks) the specimens retain a magnetization whose intensity and direction are approximately the same across the collection. In some specimens, the directions continue to change between 670 and 720°C. The carrier of this 'high-temperature' magnetization is not known; it could be hematite with blocking temperature higher than 675°C, or iron, or an iron-rich mineral such as sphene. Comparison of the palaeopole with others from rocks of about the same age from Gondwana shows a good agreement. There is an excellent agreement with a result from southern Africa (Ntonya Ring Structure) which suggests that there have been no important motions between the different blocks which form Africa, at least since that time. This pole places western Africa in high latitudes and is consistent with the presence of upper Precambrian tillites.

In order to add to the knowledge of the Late Precambrian geomagnetic Earth's field in Africa, the following study of a Pan-African dioritic intrusion from northern Mali is presented.

Geology and sampling

The Pan-African Orogeny was defined by Kennedy (1964) as a major African thermotectonic event that occurred about 550 Myr ago. Since then, deformations and intrusions of

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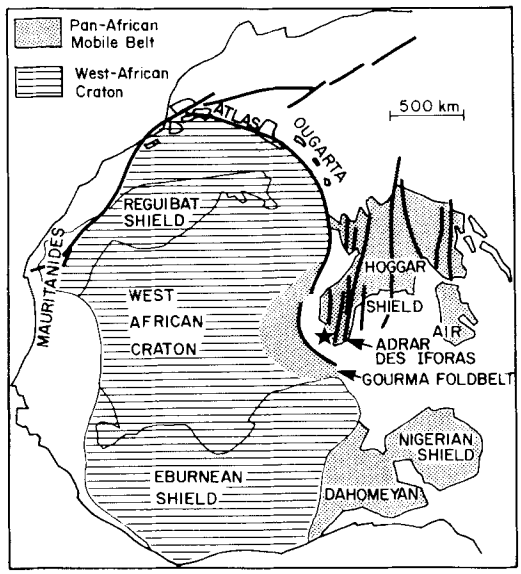


Figure 1. Structural elements of West Africa. A star shows the location of Fig. 2.

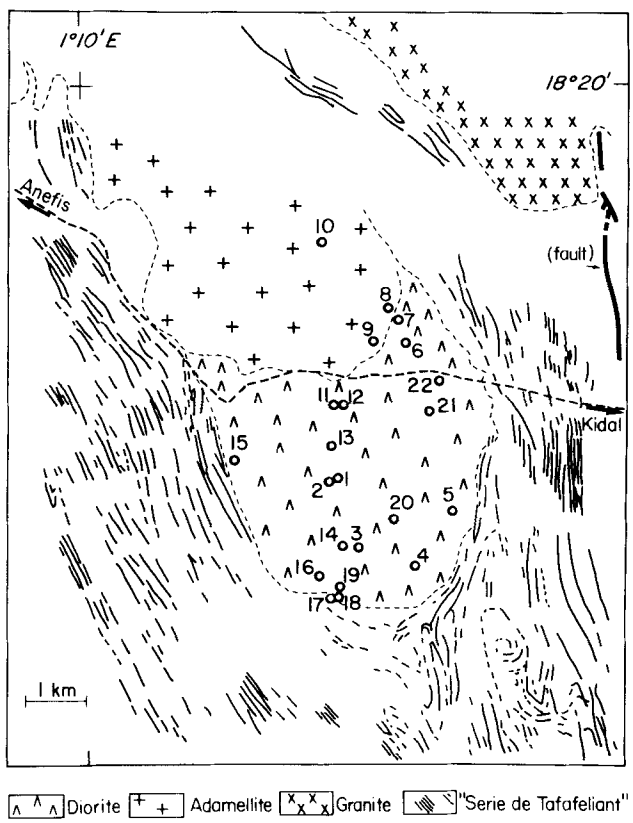


Figure 2. Sampling area and collecting sites.

this age have been recognized in South America, Australia, Antarctica and Europe. The Pan-African is represented in West Africa by the Hoggar and Nigerian shields (Fig. 1) and for a long time it was considered to be an intra-continental event. However, more recently it has been suggested that the Pan-African foldbelts of West Africa are probably caused by the approach and the collision of older nucleii (Black 1967, 1978; Caby 1970, 1978; Bayer & Lesquer 1978; Black *et al.* 1979a, b; Bertrand & Caby 1978; Bertrand *et al.* 1981).

The Adrar des Iforas of northern Mali (Fig. 1) is part of the Pharusian belt as defined by Bertrand & Caby (1978). The Adrar des Iforas is characterized by several tectono-stratigraphic units and events (Bertrand & Caby 1978; Black 1978; Boullier *et al.* 1978; Caby 1978; Bertrand *et al.* 1981; Black *et al.* 1979):

The intrusion studied (Adma diorite) is a syn- to late-tectonic Pan-African intrusion. It is composed of diorite dated at 616 ± 11 Myr, and adamellite dated at 613 ± 3 Myr (method: U/Pb on zircons (Ducrot, Lancelot & Renaud, private communication)). It intrudes the 'Série de Taféliant' as described by Fabre (1976). The Adma diorite can be taken to date the end of the Pan–African event at 620 ± 30 Myr. A few results from a post Pan-African rhyolite (Tabankort rhyolite) are also given.

Fifteen sites were drilled and seven sites hand-sampled. The sites were distributed over the intrusion (Fig. 2). Thirteen sites were taken from the diorite, three near to the contact with the 'Série de Taféliant', one (17) from the 'Série de Taféliant', itself metamorphosed at the contact by the diorite, one (14) from a post-diorite basic dyke, three (7, 8, 9) in the contact area between the diorite and the adamellite and one (10) from the adamellite itself. Usually four or five cores or hand samples were taken at each site. Orientation was obtained by both magnetic and Sun compasses. The cores (2.54 cm diameter) were cut into 2.4 cm length cylinders, and the hand samples into cubes of 3.5 cm side.

Results

The measurements were made at the geomagnetic laboratory of Parc Saint-Maur (Paris, France). Magnetizations were measured with a translation inductometer (Daly 1970). Alternating field (af) treatments were performed in fields up to 160 mT using a three-axis tumbler. Thermal treatment was performed with a furnace controlled by two thermocouples. Heating was automatic and was carried out during the night to minimize external effects. All treatments were carried out in a quasi-zero residual field (± 20 nT) provided by Helmholtz coils. Further details are given in Morel-à-l'Huissier (1978).

Natural remanent magnetizations (NRM) of 189 cylinders from the cores and 25 cubes from the hand samples were measured after the samples had first been stored for one month in quasi-zero field to eliminate the principal part of the viscous remanent magnetization (VRM). The distribution is random and it seems reasonable to relate this dispersion to lightning (Rimbert 1958) which is very common in the sampling area. The within-site magnetization intensities are variable (ranging over the intrusion from 74 700 to 36×10^{-3} A m^{-1}). An experiment of 'trainage' was carried out in the Earth's field over a period of one month. The change in the magnetization was less than 1 per cent in over 90 per cent of the samples. Evidently the natural remanence is stable on a short time-scale and essentially random in direction, but it surely does not reflect the primary magnetization of the diorite.

Except for site 2, at least one specimen per site was treated in detail by thermal or af cleaning. For sites 1, 4, 5, 6, 7, 9, 10, 11 and 22, two specimens from the same core were treated, one thermally and the other by alternating fields. For sites 5, 6, 11 and 22, the specimen treated by af was subsequently treated thermally between 670 and 720° C. A total

of 36 pilot specimens were studied. Fig. 3 illustrates the changes obtained using the different procedures.

Af analysis (Fig. 3) shows that after 5 mT most of the specimens have lost about 50 per cent of their magnetization or more; the exceptions are discussed below. The major changes in direction occur between 0 and 40 mT. After 40 mT, there is a stabilization of the magnetization in both direction and intensity. Between 40 and 80 mT (or 140 mT depending on the maximum field applied), the magnetization intensity range is 50 to 5 × 10^{-3} A m^{-1} with a mean about 35×10^{-3} A m^{-1}. The intensity after final treatment usually represents only 1 to 10 per cent of the initial value. Fig. 3 also shows that after treatment in relatively weak fields (5 to 10 mT) the specimens whose NRM was negatively inclined quickly change to a well-pronounced positive inclination.

Some specimens behave differently (Fig. 3b). Their intensity after af treatment falls to only 60–70 per cent of the initial value, but the absolute remaining intensity is of the same magnitude as that for the other specimens. The final direction is also consistent with the others. Some specimens show also a small change in direction at the beginning of the treatment prior to the stabilization in direction which remains constant up to 140 mT.

From this af study one can conclude that an important fraction of the magnetization

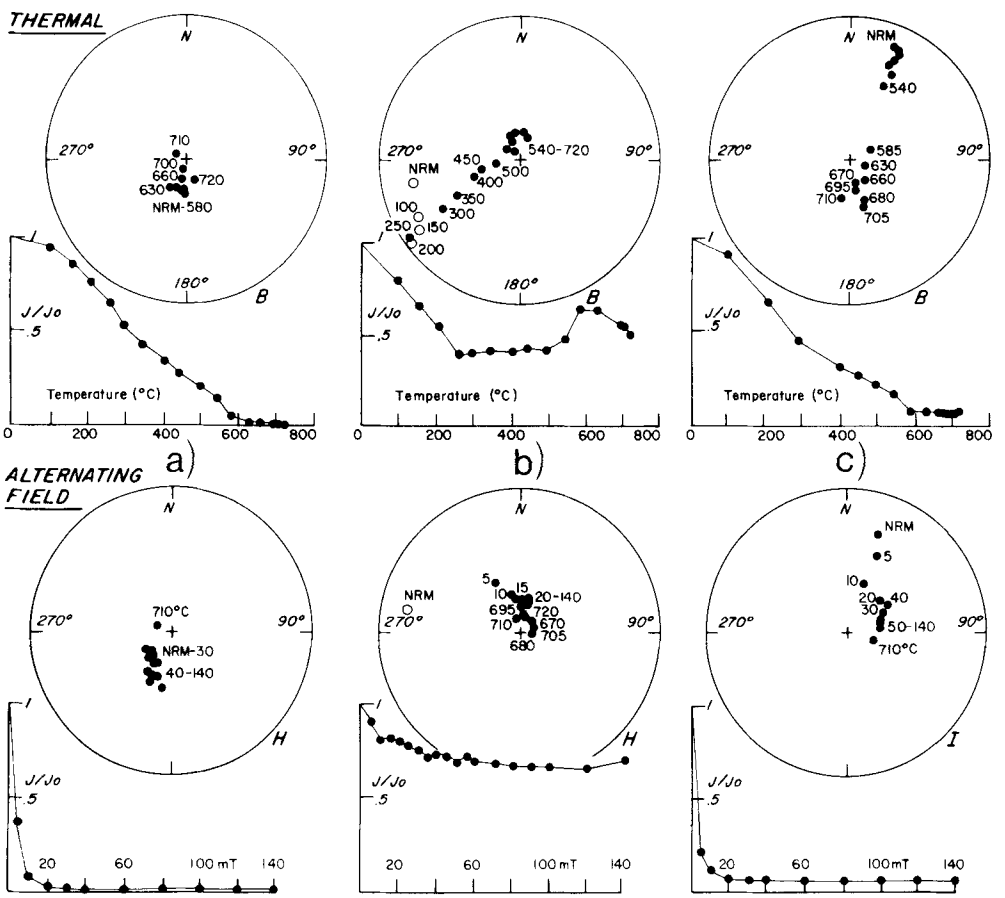


Figure 3. Thermal and alternating field treatments. (a) 10–1 adamellite, (b) 6–2 and (c) 4–4 diorite. H indicates that the specimen treated is from the top of the core, I the middle and B the bottom. Thermal results obtained after af demagnetization are indicated. Closed (open) circles indicate positive (negative) inclinations. Plane of projection is present horizontal.

('soft' magnetization) is carried by grains with weak blocking fields so that it is very sensitive to external field variations (especially to lightning) and is probably of recent origin. There is also a smaller part of the magnetization ('hard' magnetization) characterized by high blocking fields. Study of polished sections shows that the carrier of the 'soft' magnetization is probably a titanomagnetite. It occurs as large grains, more-or-less martitized, and often associated with ilmenite (Morel-à-l'Huissier 1978). The carrier of the 'hard' magnetization is probably hematite.

The grouping of the directions and the presence of good 'stable end points', might lead to satisfaction with this af treatment, but the field that created the superimposed magnetization by lightning may have affected the hematite which is the presumed carrier of the 'hard' magnetization. If it has, af demagnetization does not permit adequate separation because the peak intensity of the af used could be less than the field produced by lightning very near to the impact point.

In order to complete the analysis, thermal treatment was carried out (Fig. 3), step-by-step demagnetization up to 710 or 720°C for some specimens. The demagnetization curves show that the magnetization is carried by grains with a wide range of blocking temperature and size. First, there is a drop in intensity between 250 and 300°C, but it has not been possible to relate the magnetization removed during this interval to a component consistent in direction across the collection. The major drop occurs in the range 540–580°C and confirms the presence of magnetite (or somewhat titaniferous magnetite). Very often this drop is accompanied by a rapid change in direction. The small change in direction between 20 and 540°C, the rapid change between 540 and 580°C and the relative stable direction between 580 and 710°C (or 720°C) are shown in Fig. 3(c). Fig. 3(a, b) show two different sorts of behaviour. In 3(a) there is little change in direction between 20 and 720°C and in 3(b) there is a continuous evolution of the direction between 20 and 580°C followed by a quasi-stable direction. Generally (Fig. 3a, c), below 580°C the intensity represents 10 to 20 per cent of the initial value, and above 580°C (as for af demagnetization) the remaining magnetization represents no more than 1 to 10 per cent (average between 1 and 5 per cent) of the initial intensity. Sometimes, it is also possible to observe a comparable evolution of the direction above 580°C and up to 675°C, the Curie point of synthetic hematite. At this stage, the decrease in intensity ceases and, in the interval up to 720°C, the direction fluctuates around a mean position.

As a further test and because the af peak intensity could be lower than the intensity produced by lightning, some specimens were first treated in af fields up to 140 mT and then thermally demagnetized. Four specimens were chosen and the thermal treatment was then carried out between 670 and 720°C. Specimen 6–2 is an example (Fig. 3b). After 670°C there is a systematic small decrease in intensity sometimes accompanied by a change in direction which also indicates that the hematite may have been partially remagnetized by lightning. Another possible explanation which is partly supported by Fig. 3(b) could be that both polarities are present in some specimens. One can argue that the negative polarity is never found, but if we consider two polarities with very similar blocking temperatures or blocking fields, it becomes rather impossible, with existing techniques, to separate them. In fact, we only obtain the resultant vector which could be more-or-less near the direction of one polarity depending on the difference between the intensities of the two moments. From 680 to 720°C the direction fluctuates around a mean position and, in general, the directions found after thermal treatment are not far from those obtained after af treatment at 140 mT (which is almost stable since 40 mT).

Based on these detailed studies, the rest of the collection was treated in the following way: (1) af demagnetization at 80 mT, (2) measurement, (3) heating at 710°C, and (4)

measurement again. A total of 117 specimens (pilot specimens included) from 22 sites were treated in this manner.

These do not represent all the samples collected. Unfortunately the time available to process the specimens was limited and a choice had to be made between processing all the samples in less detail or a selection of them in greater detail. The latter procedure seemed more reasonable since the 117 specimens treated are undoubtedly representative of the intrusion as a whole.

After thermal treatment at 710°C the remaining magnetization is systematic in direction. This is unusual because the Curie temperature of synthetic hematite is ≈ 675°C. The effect is not related to poor temperature control because two thermocouples agreed to ±2°C. It is not caused by poor cancellation of the field; this was checked by carrying out duplicate heatings at the same temperature (710°C), inverting the specimens between the runs and showing that the direction after the second run remained the same. A further experiment was carried out with a specimen of synthetic hematite which was given a thermoremanent magnetization (TRM) by heating at 800°C and cooling in the Earth's field. This specimen was demagnetized along with the rock specimens. At 680°C the rock specimens retained a non-negligible magnetization, whereas the specimen of synthetic hematite had lost all its magnetization. Thus, the magnetization persisting at high temperature is not caused by experimental error and apparently has a blocking temperature between 680 and 800°C.

But what is the carrier of such a magnetization? It could be hematite. Smith & Fuller (1967) mention that the magnetization of hematite consists of a soft spin-canted moment observed in the range −20 to 675°C and a hard defect moment lost at the true Néel point of hematite (725°C). On the other hand, Park (1975, 1977) on experimental grounds notes the possibility of a higher Curie point for hematite. If the magnetization is related to hematite, it is probably not a TRM but more likely a chemical remanent magnetization (CRM), because polished sections show that hematite was formed during late-plutonic autometamorphism which probably occurred at temperatures below the Curie point of hematite. Other possibilities are native iron (this was not observed in polished sections) or sphene which can contain up to 15 per cent of iron in the form Fe_2O_3, which is abundant in these rocks and often associated with magnetic minerals (Morel-à-l'Huissier 1978).

Analysis

The results are given in Table 1. When two specimens from the same core were treated, the average of the two values was used for calculations of the overall mean values. Fig. 4 shows

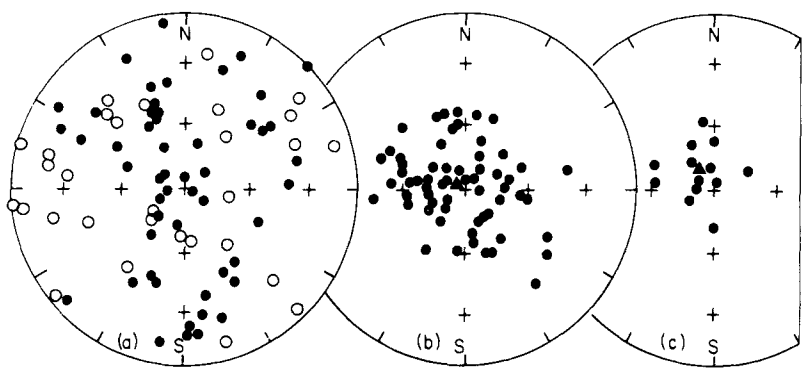


Figure 4. Directions of magnetization. (a) NRM, (b) cores and (c) sites after treatment. The mean is indicated by a triangle. Convention as in Fig. 3.

Table 1. Mean direction and VGP for each site.

Site	C	S	$D(^\circ)$,	$I(^\circ)$	k	α_{95}	Lat. (°),	Long. (°E)
1	5/5	6	278,	61	114	6	18 N,	311
2	4/2	2	Too Few					
3	6/5	6	247,	77	17	15	08 N,	339
4	5/5	6	181,	72	6	26	15 S,	001
5	5/5	6	320,	73	22	13	41 N,	335
6	5/4	6	021,	88	26	14	22 N,	003
7	6/6	8	278,	81	14	15	20 N,	343
8	5/3	6	000,	68	10	26	58 N,	001
9	5/4	6	332,	66	6	28	53 N,	330
10	5/4	7	343,	80	13	20	37 N,	354
11	5/5	7	308,	81	21	14	29 N,	345
12	4/4	4	350,	58	51	10	68 N,	341
13	5/4	7	062,	72	8	29	30 N,	035
14[1]	5/5	5	152,	−23	3	49		
15	4/4	6	Random					
16	5/5	6	212,	−13	4	30		
17[2]	5/5	5	141,	−30	3	52		
18[3]	3/3	3	181,	−39	7	52		
19[4]	5/2	3	Too Few					
20	4/4	5	290,	60	10	23	27 N,	308
21	3/3	3	Random					
22	3/2	3	Too Few					

Notes:

C drilled/used cores; S specimens; D° declination; I° inclination; k precision; α_{95} semi-angle of the cone of confidence; Lat.° latitude of the palaeopole; Long.° longitude of the palaeopole. Each specimen was at least demagnetized by af at 80 mT and heated at 710°C. Some of the pilot specimens were treated by af at 140 mT and others were heated at 720°C. (1) Basic dyke post-diorite; (2) 'Série de Taféliant' at contact; (3) diorite at contact; (4) diorite at about 100 m from the contact.

the results obtained for the 13 sites used in the final calculations (sites 1, 3 to 13 and 20). Results from nine sites have not been used for the following reasons: site 14 is a post-diorite dyke; sites 15, 21 and 22 have internally random distribution of directions even after magnetic and thermal cleaning; the directions at sites 2 and 19 agree with the direction at the 13 sites used but have not been included because they are based on two specimens only; site 16 gives a very different mean direction; sites 17 and 18 sampled at the 'Série de Taféliant' – diorite contact give also a different mean direction. It is noteworthy that site 19 sampled 100 m from 17 and 18 give a direction ($D = 137^\circ$, $I = 53^\circ$) that is consistent with the mean direction. The argument of the presence of two polarities can also be used to explain the dispersion within sites 15, 16, 17, 18, 21 and 22. At these sites, both polarities could be present with a more-or-less equal intensity, and if these polarities are slightly misaligned (because of secular variation), they could produce a nearly horizontal resultant vector (Table 1). Fig. 4(a) shows the NRM of all the specimens treated from the 13 sites. Fig. 4(b, c) show the results from each core (results from sites 2 and 19 included) and from each site. The effect of cleaning plus the averaging of specimens from the same core is seen by comparing Fig. 4(a, b). The effect of averaging from cores to sites is seen by comparing Fig. 4(b, c). The mean of all cores ($D = 301.7^\circ$, $I = 86.1^\circ$, $N = 62$, $\alpha_{95} = 6.1^\circ$) is not significantly different from the mean ($D = 317.3^\circ$, $I = 78.6^\circ$, $N = 13$, $\alpha_{95} = 9.0^\circ$) calculated from

Table 2. Mean directions and palaeopoles.

	N	D	I	k	α_{95}	Lat. N	Long. E	d_m	d_p
(1) Within specimens	62	301.7	86.1	9	6.1	22.2	354.1	12.2	12.0
(2) Within sites	13	317.3	78.6	22	9.0	33.5	343.5	17.1	16.1
(3) Sites as VGPs	13	–	–	–	–	32.5	344.7	15.9★	

★ Value of A_{95}.
All these values are after af treatment at 80 mT followed by thermal treatment at 710°C.

the sites (Fig. 4c and Table 2). In Table 2, the palaeopole is calculated in three different ways from the mean of cores, from the mean of sites and by averaging the virtual geomagnetic poles (VGPs) for each site. The results are very close. They are also in excellent agreement with the mean calculated after af treatment at 80 mT ($D = 311.2°$, $I = 80.5°$, $N = 51$, $\alpha_{95} = 7.1°$ giving cores unit weight; $D = 319.8°$, $I = 77.5°$, $N = 11$, $\alpha_{95} = 10.8°$ giving sites unit weight). As the intrusion did not cool instantaneously, but rather over a long period, it seems reasonable to assume that the magnetization at each site is uniform and records the geomagnetic field for a given time that is short compared to the cooling interval spanned by all the sites from the intrusion. Therefore, the preferred summary value is line 3 of Table 2 in which the mean of the VGPs has been obtained by treating the specimens first at 80 mT and secondly at 710°C.

It seems probable that the stable high-temperature magnetization is caused by hematite which is a product of chemical changes during late plutonic autometamorphism. If this is correct, the time at which this process ceased will be the age of the magnetization. The time at which this alteration ceased is probably between the time of intrusion of the diorite (616 ± 11 Myr) and the intrusion of the late stage ring complexes (590 Myr, Ducrot *et al.*, private communication). The latter are the final igneous events in this region and mark the time when the region finally cooled after the Pan-African Orogeny. According to this argument, the magnetization is in the age limits 616–590 Myr, and this seems to be the most reasonable choice.

Preliminary results from the Tabankort rhyolite support this opinion. The Tabankort rhyolite (19.7° N, 1.1° E) made up of a succession of green and red flows is also from the Adrar des Iforas but north of the Adma diorite. The rhyolite overlies the intrusive suite of which the Adma diorite is a part and which is considered to be equivalent to some nearby granitic intrusions that yield a zircon age of about 590 Myr (Ducrot *et al.*, private communication). This preliminary study is based on eight specimens from three sub-horizontal lava flows. Af demagnetization up to 140 mT and thermal demagnetization up to 715°C were done. These specimens give a mean direction $D = 114°$, $I = -70°$ ($\alpha_{95} = 7°$) and the corresponding palaeopole is 30° N, 322° E ($dm = 12°$, $dp = 10°$). This palaeopole is in good agreement with that from the diorite, but the direction is reversed. This agreement between such different rock types is good evidence in support of the conclusion that the age of the palaeopole from the Adma diorite is about 590 Myr.

Discussion

Fig. 5 shows other palaeopoles from Gondwana from rocks of similar age to the Adma diorite. There is a good agreement between these poles, and specially notable is the agreement between the Adma diorite palaeopole (AD) and the Ntonya Ring Structure palaeopole (NRS) the latter body being dated at 630 ± 24 Myr. This shows for the first time from well-

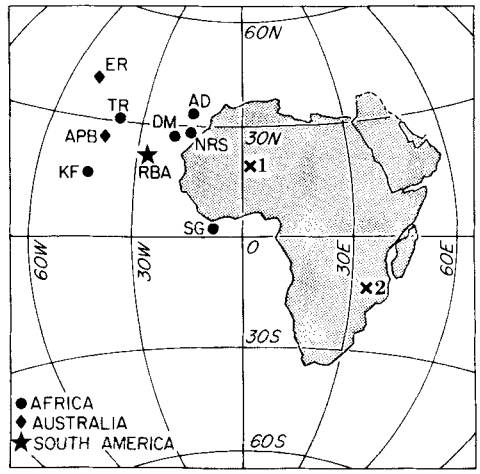


Figure 5. Late Precambrian–Lower Cambrian palaeopoles for Gondwana. NRS Ntonya Ring Structure; SG Sijarira Group; DM Deza Mountain; KF Klipheuvel Formation; AD Adma diorite; TR Tabankort rhyolite; ER Edith River; APB Antrim Plateau basalts; RBA red beds of Argentina. ER and APB are corrected using the reconstruction of Smith & Hallam (1970); RBA is corrected using the reconstruction of Bullard, Everett & Smith (1965). 1 and 2 are respectively the sampling localities of AD and NRS.

dated rock units that the southern and the western part of Africa have probably kept their actual relative position at least since this period. But, because of its post-tectonic origin, this result does not corroborate or nullify the hypothesis that the Pan-African foldbelt is a collision Orogeny. Perhaps, as the systematic dating program of the Adrar des Iforas proceeds, it will be possible to find a suitable older material (≃ 700 Myr) to study this question by palaeomagnetic means.

If we consider the Tabankort rhyolite, it seems that the pole moves westward from the west African coast. This supports the hypothesis of McElhinny, Giddings & Embleton (1974) and Veevers & McElhinny (1976) of a polar migration from south-eastern Gondwana to north-western Gondwana, placing Africa during the Late Precambrian in high latitudes

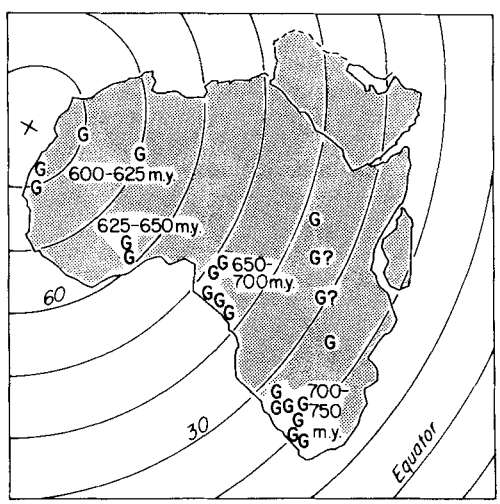


Figure 6. Palaeolatitude map for Africa. G occurrences for late Precambrian glaciation, their approximate ages are from Kröner (1977) and given in 10^6 yr.

(Fig. 6) in opposition to Piper's (1973) conclusions. Palaeolatitudes of Fig. 6 have been derived from the Adma diorite palaeopole and do not necessarily correlate with the oldest glacial occurrences shown in the figure. The results presented here tell us nothing about the problem of the non-synchroneity (Kröner 1977) or the diachronism (Deynoux *et al.* 1978) of Late Precambrian glaciations, but it is interesting to mention that the proposed correlation of Kröner (1977, p. 293) for the last glaciation of the Late Precambrian seems to be in excellent agreement with a polar migration ($\simeq$ 725 Myr for the Numees Formation and Chuos Formation of southern Africa to $\simeq$ 600 Myr for the Taoudeni basin, Mauritania, Fig. 6). Though this palaeopole needs corroboration by other African studies, it disagrees with the palaeomagnetic reconstruction proposed by Morel & Irving (1978) for the Precambrian–Cambrian boundary. That reconstruction was in good agreement with Piper's (1973) interpretation which has been supported by Williams (1974, 1975) and discussed by Kröner (1977). The existence or non-existence of Gondwana prior to Late Precambrian times cannot be established from the palaeomagnetic data presently available. This is mainly because of the poor age control of most of the palaeopoles, especially for those from Australia and South America.

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